pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Two Series: Cointegration and Mean Reversion

WEEK 2 · DAY 4

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Today: six steps

Six steps

1. Levels against changes: two correlations
2. The hedge ratio and the gap between the series
3. Engle-Granger, every setting of coint
4. The half-life of the gap, with the sign checked
5. A rolling ratio, and the next fold
6. What the numbers describe, and what they do not

Two pairs

The 5 Yr and 10 Yr yields (real, public file) and the invented IG and HY spreads (Course 3's, not any index). Development years only: 2 Jan 2015 to 30 Dec 2022, 2,001 dates.

A description of past dates. Never a trade.

Names for a pair

The words for a pair of series. What this day says about one:

Allowed names

the gap between the two series

half-life in trading days

hedge ratio (a fitted slope)

Never called

a trade, an entry, or a signal

rich or cheap

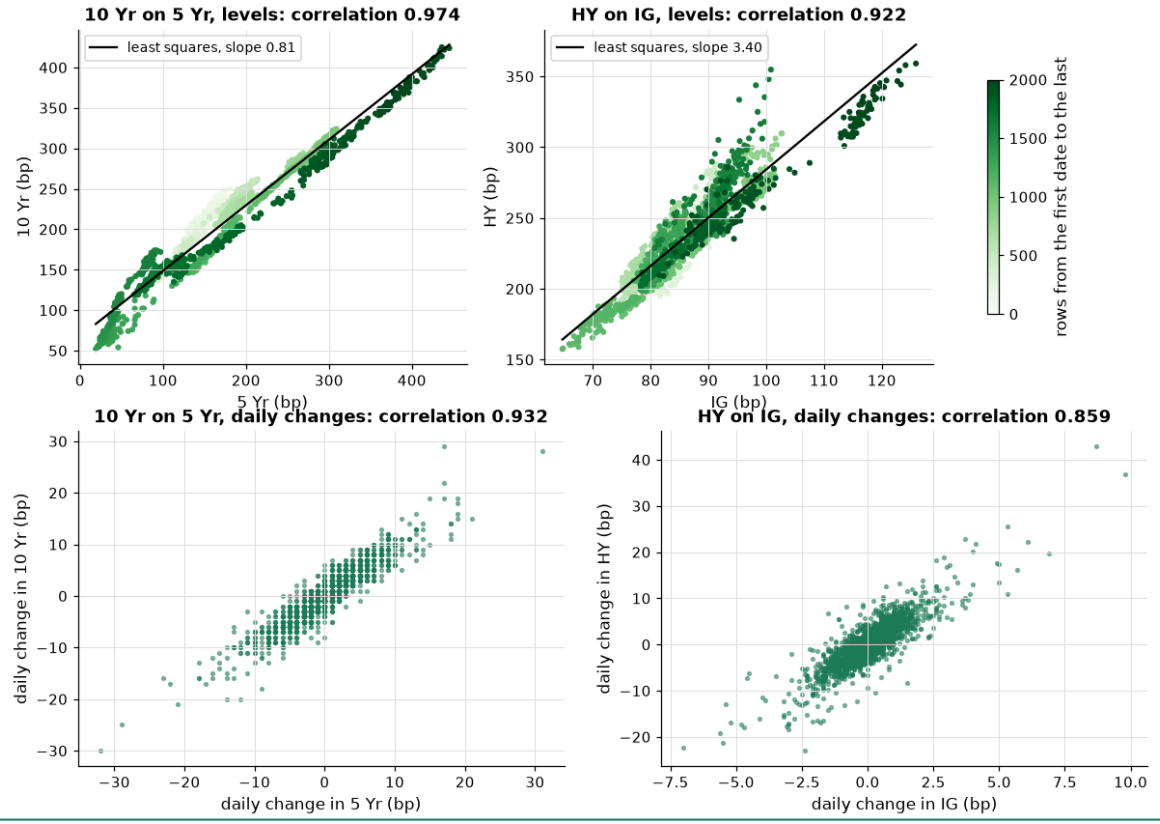
a forecast that the gap closes

The gap between the two series closed at a half-life of N trading days over these years.

How levels against changes works

Correlation	How closely two series move together, from -1 to 1
On levels	A poor guide: two series that each wander look related whatever ties them
On changes	The wandering is gone: do the two move together day to day?
The limit	Neither says there is a stable relationship in the levels
What comes out	Two numbers per pair, to be read together

Levels against changes



Pair	Levels	Changes
10 Yr on 5 Yr	0.9741	0.9316
HY on IG	0.9222	0.8590

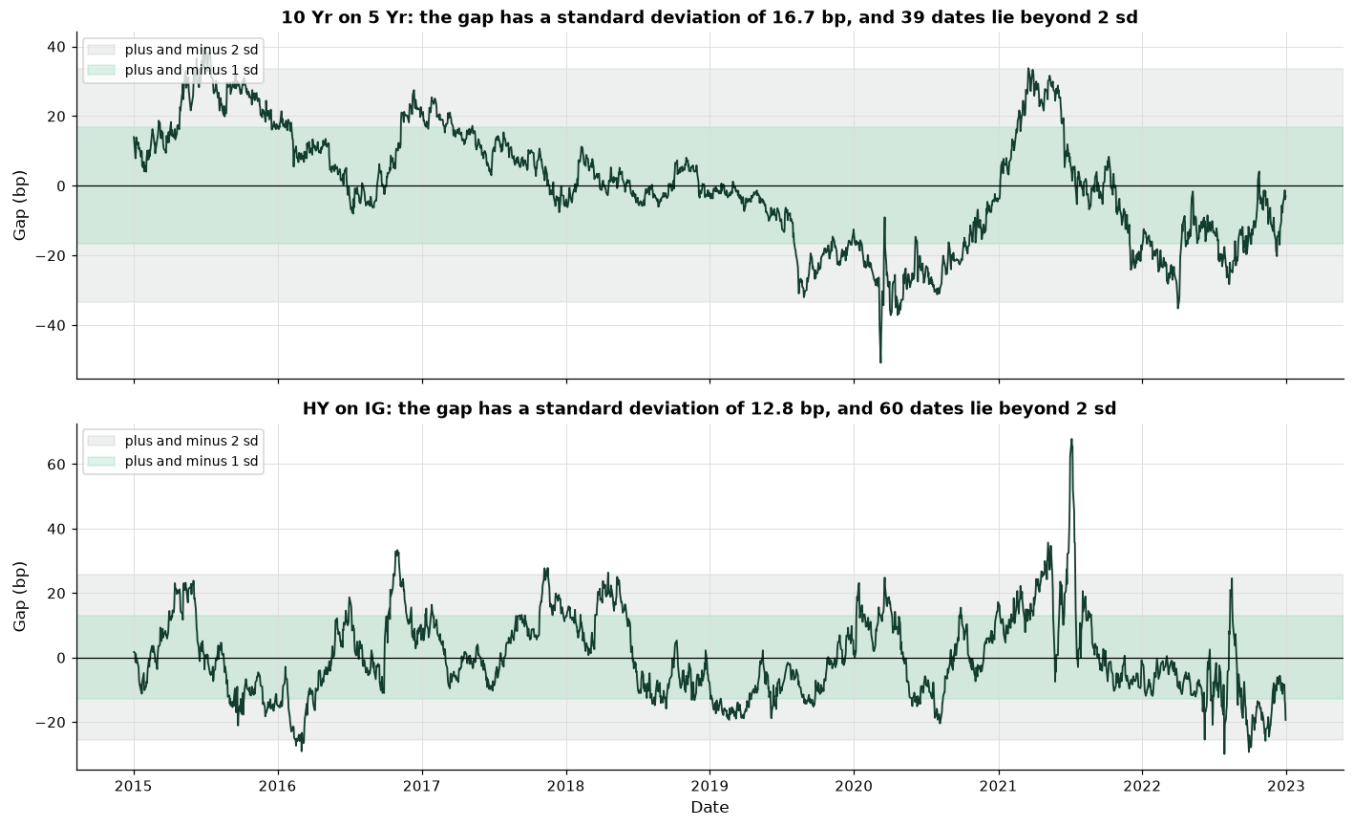
Both pairs: highly correlated in levels, and in changes. The points drift along the line in a path, not a cloud: high correlation does not decide cointegration.

Correlation of the levels and of the daily changes, development years.

How the hedge ratio works

The line	Least squares of y on x: y = constant + ratio x
The ratio	The bp y moved for each bp x moved, on average over the dates fitted
The gap	y less the line: positive above it, negative below, average zero
The question	Does the gap keep returning to its average? That is cointegration
What comes out	The constant, the ratio, and the gap on every date

The gap between the series



Pair	Ratio	Const	Sd	>2 sd
10 Yr on 5 Yr	0.810	67.6	16.7	39
HY on IG	3.402	-55.9	12.8	60

The bands describe how big the gap was, in units of its own spread. They are not a rule for anything. Look at the shape: the first gap wanders; the second crosses zero more often.

Ratio, constant (bp), standard deviation of the gap (bp), and dates beyond 2 sd.

How Engle-Granger works

Cointegrated	Some mix of the two is stationary, though each alone is not
Two steps	Fit the line of y on x; test the gap for a unit root
The null	No cointegration: a small p-value rejects it
The catch	The gap was fitted, so the ordinary ADF critical values are too lenient; MacKinnon's are used
What comes out	A statistic, a p-value, and three critical values

Engle-Granger on both pairs

Pair	t	p-value	5% crit.	ordinary ADF p
10 Yr on 5 Yr	-2.1928	0.4281	-3.3392	0.2091
5 Yr on 10 Yr	-1.9594	0.5496	-3.3392	0.3056
HY on IG	-4.2353	0.0032	-3.3392	0.0006
IG on HY	-3.8249	0.0126	-3.3392	0.0027

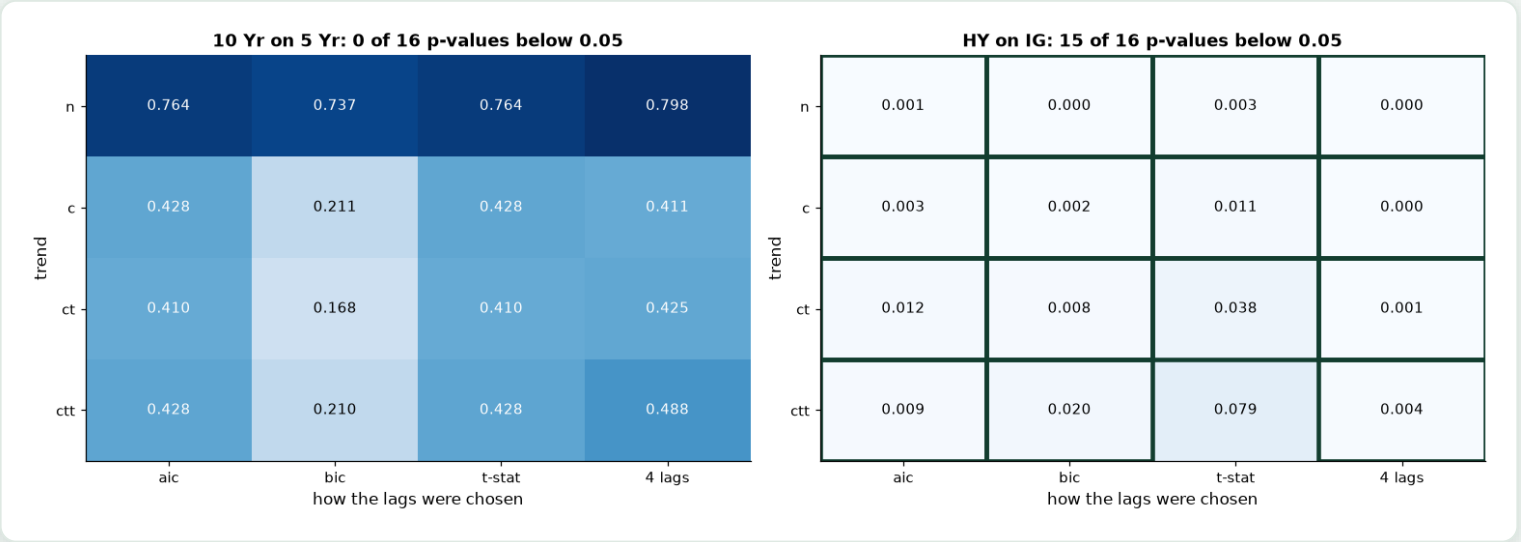
Rejects no cointegration at 5 percent for: HY on IG, IG on HY. The 10 Yr on the 5 Yr: not rejected, however correlated.

The ordinary ADF p-value is smaller than the proper one in both pairs: the trap of testing a fitted gap with day 3's critical values.

Use coint. The order of y and x changes the regression, and the p-value a little.

Every setting of coint

Setting	Default	Tune first
y0, y1	required	yes
trend	'c'	yes
method	'aeg'	no
maxlag	None	no
autolag	'aic'	yes



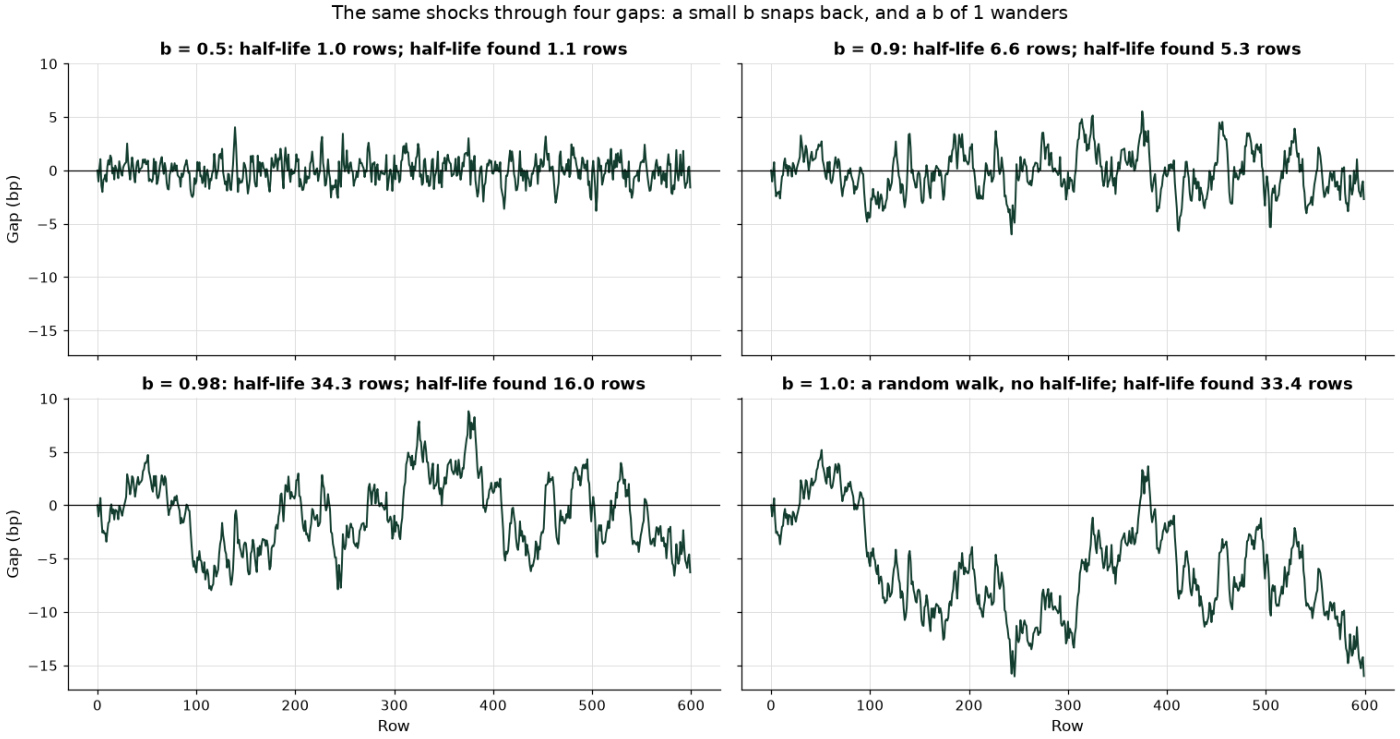
p-values below 0.05 of 16: 10 Yr on 5 Yr 0, HY on IG 15. A conclusion that survives every reasonable setting is worth more than one that needs a particular setting.

The defaults are read from the installed statsmodels with inspect.signature, never typed.

How the half-life works

The model	An AR(1) on the gap: each value is a constant plus b times the value before, plus a shock
The pull	The distance from its level shrinks by b every row
The half-life	ln(0.5) / ln(b) rows until the distance has halved
The sign check	b must be above 0 and below 1; otherwise there is no half-life
What comes out	b, and a number of rows

The half-lives



Pair	b	Half-life	EG p
10 Yr on 5 Yr	0.9931	99.5919	0.4281
HY on IG	0.9804	34.9771	0.0032

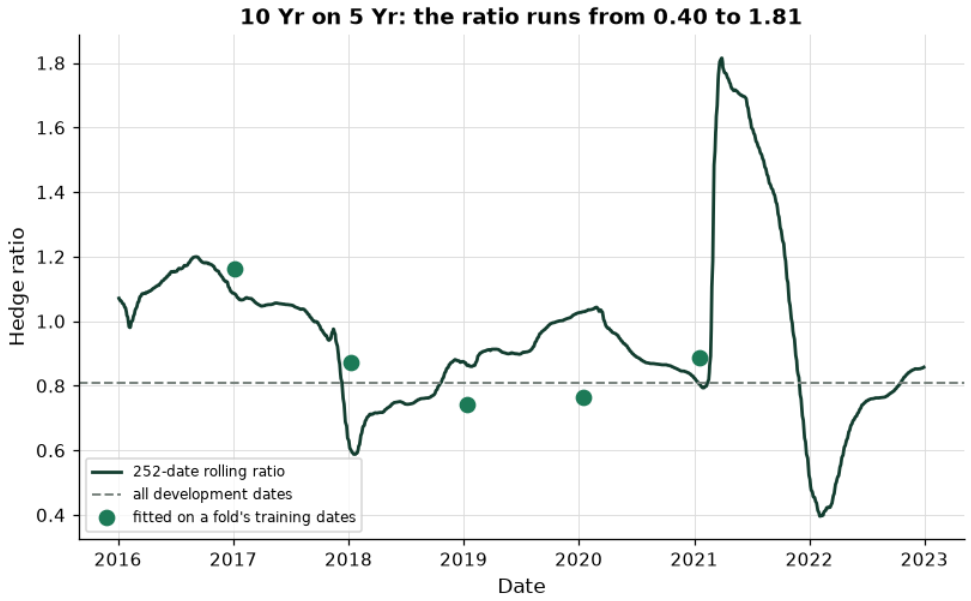
A half-life is not a test: the first pair has a b below 1 and was not found cointegrated. The sign check: half_life refused a gap that grows 2 percent a row, because its b is above 1.

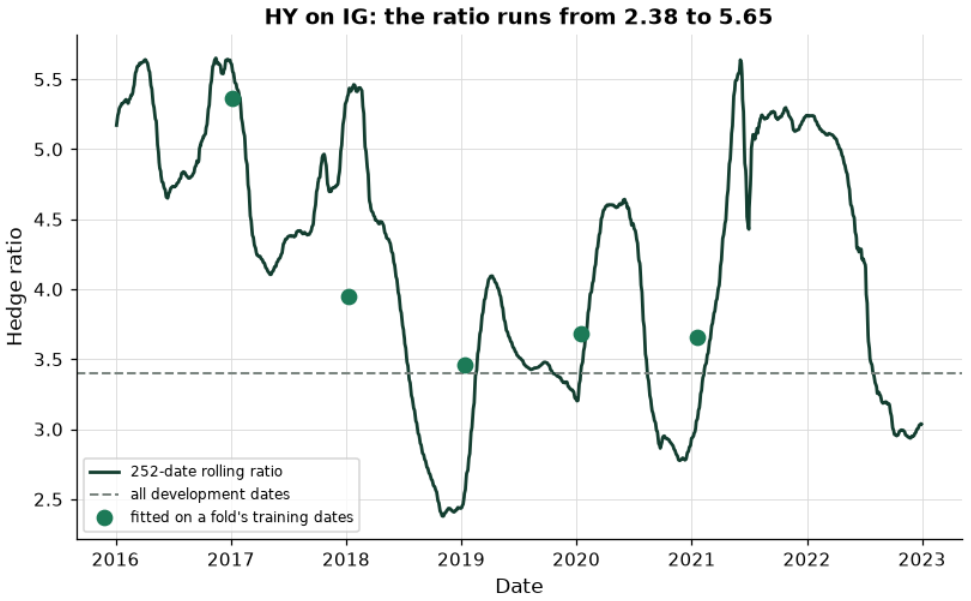
A half-life from a few hundred rows is a rough number: a random walk gave one.

How the rolling and fold views work

The worry	A relationship is only worth describing if it holds on dates the fit did not see
Rolling	Refit the line on the last 252 dates, on every date: a ratio over time
The fold view	Fit on a fold's training dates; build the gap on the next dates with that fit
The question	Does the next gap still return to a level? Its b, and its half-life if it has one
What comes out	A path, and a table by fold

The ratio over time

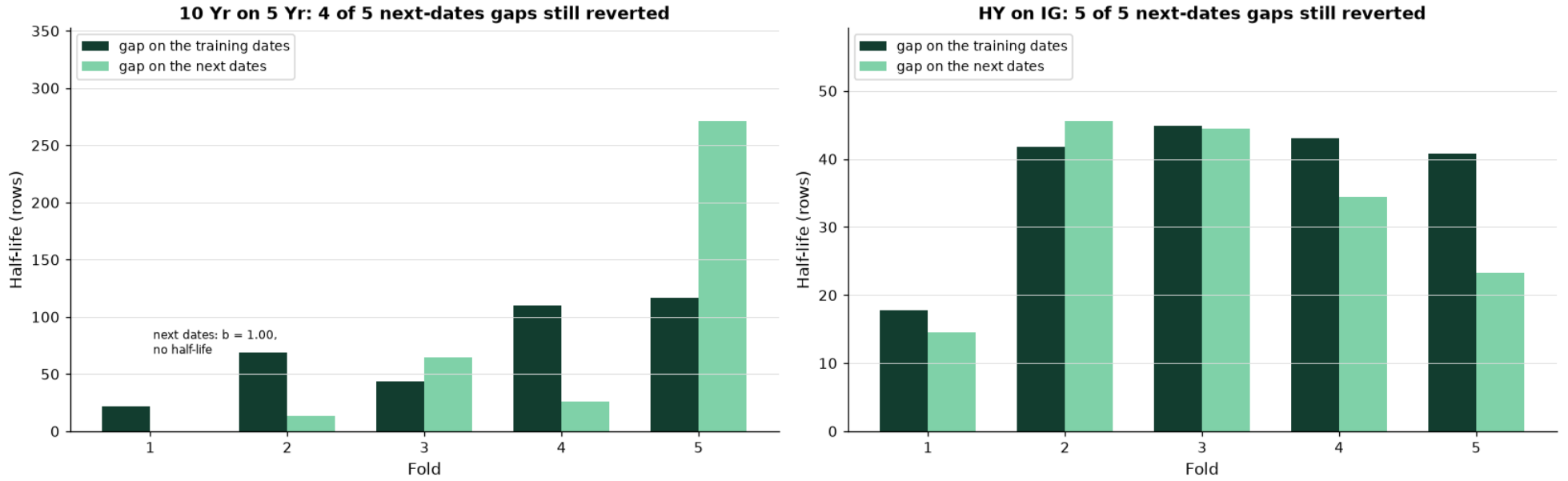




The ratio is a path, not a constant. A hedge ratio fitted on one stretch is a description of that stretch; applied to the next, it is a different line from the one the next stretch would give.

The window includes the date: it describes how the relationship stood, and scores nothing.

The next fold



The gap on the next dates still reverted in 5 of 5 folds for the HY on the IG, and 4 of 5 for the 10 Yr on the 5 Yr; in the fold where it did not, the slope was 1 or above.

The walk-forward version of the question, on day 2's folds.

What these numbers describe, and what they do not

What they describe

Over the development years the 5 Yr and 10 Yr moved together and the test did not find them cointegrated. The gap between the invented HY and IG spreads closed at a half-life of N trading days over these years, and it reverted on the next dates in most folds.

What they do not

Not a forecast that a gap will close

Not a view that a gap is wide or narrow

The bands describe size, not an action

The relationship changed between stretches

The invented spreads were generated tied together

A description of past dates inside the file: not a view on rates, and not a trade.

Excel side by side

Excel side by side (tested in Excel)

`=SLOPE(C2:C2002,B2:B2002)`	ratio 0.8104
`=C2-($M$2+$M$1*B2)`	gap sd 16.72 bp
`=SLOPE(F3:F2002,F2:F2001)`	b 0.993064
`=LN(0.5)/LN(M4)`	99.59 rows
`=CORREL(B2:B2002,C2:C2002)`	levels 0.9741
`=CORREL(H3:H2002,I3:I2002)`	changes 0.9316

10 Yr on 5 Yr; the HY on the IG half-life is 35.0 rows. Excel has no ADF or Engle-Granger test.

Save the day: searching the history

1. Commit

The code, then the two hedge ratios

2. Grow

Add the half-lives to the same file

3. Search

git log -S the number, -- the file

4. Read

Only the second commit lists it

```
Day 9: add the half-lives
Day 9: hedge ratios
Day 9: tskit with hedge_ratio, half_life, and rolling_hedge_ratio
```

The search text is 35.0, the HY on IG half-life.

A line of results is something you can ask Git about.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the slope's sign is wrong: the function returns a half-life of -6.0 rows for a gap whose half-life is positive.

Code that runs is not the same as code that is right.

What you can do now

- Levels against changes: correlation does not decide cointegration
- A hedge ratio, and the gap between the series
- Engle-Granger with coint: the ordinary ADF p-value is too lenient
- A half-life from an AR(1), with the sign checked
- The ratio is a path: the next fold is the walk-forward question

Week 2, Day 5 Course 5B capstone